Adiba Fairooz Chowdhury

B.Sc. in Computer Science & Engineering, Chittagong University of Engineering & Technology (CUET)

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Website | Google Scholar | ResearchGate

Research Interests: Robust & Secure LLMs (adversarial robustness, trustworthy AI), Multimodal and Low-Resource NLP, Explainable AI, with applications in Healthcare

Education

•Chittagong University of Engineering & Technology (CUET) *2022–2026*

B.Sc. in Computer Science & Engineering | CGPA: 3.68/4.00

- **Undergraduate Thesis:** *A Progressive Adversarial Benchmark and Cross-Modal Consistency Guided Detection for Bengali Multimodal Fake News Detection*
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Digital Image Processing, Digital Signal Processing, Information Security, Algorithms, Applied Statistics, Linear Algebra, Mathematical Programming

•Government Hazi Muhammad Mohsin College *2018–2020*

Higher Secondary Certificate (HSC) | GPA: 5.00/5.00

Research Publications

C = Conference | J = Journal | S = In Submission | T = Thesis

[J1] M. S. Chowdhury, **A. F. Chowdhury**, A. Banu, R. Hossain, and M. Chowdhury, "Automated Resume Evaluation Using Large Language Models: A Multi-Agent Framework with Fine-Tuning and Prompt Engineering," *IEEE Access*, vol. 14, pp. 84085–84102, 2026. DOI.

[J2] M. S. Chowdhury, **A. F. Chowdhury**, A. Banu, and R. Hossain, "CareerCorpus: A Comprehensive Dataset of Annotated Resumes," *Data in Brief*, Elsevier, 2026. DOI.

[C1] M. S. Chowdhury, **A. F. Chowdhury**, A. Banu, and R. Hossain, "Automated Bengali Physics MCQ Solver: A Large Language Model Approach with Chain-of-Thought Reasoning," in *Proc. IEEE QPAIN*, Chattogram, Bangladesh, 2026. DOI.

[C2] **A. F. Chowdhury** and M. S. Chowdhury, "Quasar@EEUCA 2026: Multimodal Deep Learning for Vaccine Stance Detection in Memes," in *Proc. 9th Workshop on Event Extraction and Understanding (EEUCA)*, ACL, San Diego, USA, 2026. DOI.

[C3] M. S. Chowdhury and **A. F. Chowdhury**, "Quasar at MAHED Shared Task: Decoding Emotions and Offense in Arabic Text using LLM and Transformer-Based Approaches," in *Proc. ArabicNLP*, ACL, Suzhou, China, 2025.

[C4] **A. F. Chowdhury** and M. S. Chowdhury, "Quasar at AraHealthQA Track 1: Leveraging Zero-Shot LLMs for Q&A Categorization in Arabic Mental Health," in *Proc. ArabicNLP*, ACL, Suzhou, China, 2025.

[P1] M. S. Chowdhury and **A. F. Chowdhury**, "Robust Long-Form Bangla Speech Processing: ASR and Speaker Diarization," *arXiv preprint* arXiv:2602.21741, 2026.

[S1] M. S. Chowdhury, **A. F. Chowdhury**, and M. Chowdhury, "Myocardial Infarction Prediction Using Adaptive Machine Learning and Explainable AI Techniques," *Discover Artificial Intelligence*, 2026. *Under Revision.*

[C5] M. Sultana, M. S. Chowdhury, T. A. Ridi, **A. F. Chowdhury**, W. Alam, and F. Amin, "Elegant: Skin and Hair Care Assistant," in *Proc. IEEE ICCCT*, Chennai, India, 2025. *RA Contribution: manuscript editing and critical review.*

[C6] M. A. Ashir, M. S. Chowdhury, **A. F. Chowdhury**, M. Chowdhury, R. Aich, and N. Das, "An AI-Enhanced Four-Class Stacking Ensemble Framework for Explainable Income Tax Fraud Detection Using Calibrated NBR IT-10B Structure," in *Proc. IEEE OMLET*, 2026. *Accepted. RA Contribution: manuscript drafting and review, related-work analysis, methodology and algorithm design, experimental execution, and results analysis.*

[C7] H. Akter, M. Chowdhury, M. S. Chowdhury, and **A. F. Chowdhury**, "Multiclass Pregnancy Risk Prediction Using Feature Engineering, Ensemble Learning, and Explainable AI," in *Proc. IEEE OMLET*, 2026. *Accepted. RA Contribution: manuscript drafting, exploratory dataset analysis, methodology design, and research guidance throughout study development.*

[C8] A. S. Dipto, M. S. Chowdhury, **A. F. Chowdhury**, R. Dhar, A. Rahman, and M. Chowdhury, "An Explainable Multi-Class Machine Learning Framework for Differential Diagnosis of Mosquito-Borne Diseases," in *Proc. IEEE OMLET*, 2026. *Accepted. RA Contribution: literature review, methodology development, experimental execution, data analysis, and manuscript writing.*

[C9] N. Shahriar, M. S. Chowdhury, **A. F. Chowdhury**, and M. Chowdhury, "URL-Only Phishing Detection Based on Structural and Character-Level Lexical Fusion Using TF-IDF and LightGBM," in *Proc. IEEE i-COSTE*, 2026. *Accepted. RA Contribution: manuscript drafting and review, literature analysis, methodology development, experimental execution, and results analysis.*

Research Experience

•ELITE Research Lab (ELITELab AI) *Jun 2026 – Present*

Research Assistant Remote

– Developing an AI-powered educational platform for Bangladeshi students.

– Researching robust few-shot adaptation of the Segment Anything Model for four-chamber heart segmentation under challenging image conditions, including blurry and grainy echocardiographic scans.
– Conducting literature review, methodology design, implementation, experimentation, and analysis for AI/ML research projects, contributing to research publications and open-source systems.

•SoftRobotics *Nov 2025*
Industrial Attachment – Full Stack Development Chittagong
– Developed a Laravel payment gateway system through collaborative development using GitHub.
– Implemented payment processing, refund mechanisms, and multiple payment gateway integrations.
– Gained production experience with Laravel, PHP, and MySQL in a professional environment.

Selected Research Projects

•Automated Resume Evaluation Using Large Language Models – Multi-Agent Framework 🔗 *IEEE Access | 2026*
Python, Qwen, Gemma, LLM Fine-Tuning, Multi-Agent Systems, Prompt Engineering

– Designed a heterogeneous multi-agent architecture – specialized Education, Skills, and Experience agents built on fine-tuned LLMs – that decomposes a complex evaluation task and aggregates agent outputs for candidate-job matching
– Achieved MAE 0.065, Pearson r 0.896, R^2 0.766, outperforming single-model baselines (GPT-5, Gemini, Claude, Grok, Mistral)

•Quasar at AraHealthQA & MAHED – Cross-Lingual LLM Transfer for Arabic NLP 🔗 *ACL Workshops (ArabicNLP 2025)*
Python, Qwen3-14B, Gemma-7B, XLM-RoBERTa, ArabERT, Zero-Shot Inference

– Studied zero-shot transfer of general-purpose LLMs to a low-resource language (Arabic) without task-specific fine-tuning; 4th place on mental-health Q&A categorization (Question F1 0.52, Answer F1 0.76) via targeted prompt design
– Benchmarked transformer and LLM approaches for Arabic hate/hope speech & emotion detection – 6th place (Macro-F1 0.48) and 12th place (Macro-F1 0.674) across two subtasks

•Robust Long-Form Bangla Speech Processing – Low-Resource ASR & Diarization 🔗 *DL Sprint 4.0, BUET CSE Fest | 2026*
Python, PyTorch, HuggingFace Transformers, Demucs, pyannote, Whisper

– Finalist – top 17 of 192 teams. Adapted a pretrained multilingual speech pipeline (Demucs separation, silence-aware chunking, BengaliAI Whisper) to long-form, low-resource Bengali audio, cutting WER to 0.377 (∼47% reduction from baseline)
– Fine-tuned a pretrained pyannote segmentation model on limited Bengali data, improving DER by 12.41 absolute points to 0.277

•Quasar@EEUCA 2026 – Multimodal Ensemble for Vaccine Stance Detection 🔗 *ACL Workshop (EEUCA 2026) | 2026*
Python, PyTorch, DeBERTa-v3, RoBERTa, CLIP, OpenCV

– Combined text and vision encoders (DeBERTa-v3-large, RoBERTa-large, CLIP ViT-B/32) via weighted soft voting to reason over noisy, multimodal meme data; Macro-F1 0.8306, ranked 8th of 25 international teams

•Bangla Physics MCQ Solver 🔗 *CUET Intra ML Contest | 2025*
Python, HuggingFace Transformers, Qwen2.5, Few-Shot Learning

– 3rd place in Phase I; implemented few-shot learning with Qwen2.5 for Bengali physics comprehension (90% public / 84% private accuracy)

Technical Skills

ML / AI: PyTorch, HuggingFace Transformers, Scikit-learn, LoRA/PEFT, NLP, Prompt Engineering, Zero/Few-Shot Learning, LLM Fine-Tuning
Languages: Python, C/C++, Java, JavaScript/TypeScript, PHP, SQL
Tools: Git, GitHub, Jupyter Notebook, LaTeX, Postman
Web & Databases: React.js, Node.js, Laravel, .NET Core, MongoDB, MySQL, SQL Server

Achievements

•BUET CSE Fest DL Sprint 4.0 (ASR & Diarization) Finalist | Top 17 out of 192 teams *2026*
•AraHealthQA Shared Task 4th Place – Zero-Shot Q&A Categorization in Arabic Mental Health *2025*
•CUET Intra ML Contest 3rd Place in Phase I – Bangla Physics MCQ Solver *2025*
•Shobdotori Multi-Dialectal Bengali Speech Recognition 30th Place – ASR using Transformers *2025*
•Academic Scholarship ×7 – Chittagong University of Engineering & Technology *2022 – 2026*

Certifications

•Supervised Machine Learning: Regression and Classification– Coursera
•.NET Development– Udemy
•React Frontend Development– EDGE Program (ICT Division, Bangladesh)

Other Technical Projects

Full-stack development work (Industrial Attachment at SoftRobotics; academic & personal projects): Laravel payment gateway system, MERN-based photo exhibition platform, .NET/React inventory management system, and a Chrome extension for academic result tracking. github.com/AdibAFC